

Infection-Related Glomerulonephritis

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KEY POINTS

The umbrella term “infection-related glomerulonephritis (IRGN)” includes glomerular diseases caused by bacteria, viruses, fungi, or parasites. While bacteria are commonly associated with IRGNs in industrialized countries, chronic parasitic infections may play a greater role worldwide.

A standardized nomenclature of IRGN is lacking. Various terms are used for overlapping entities defined by the causative pathogen, site of infection, temporal relationship of the glomerular disease to the infection, and histologic characteristics.

The epidemiology of IRGN has significantly changed over recent decades. The incidence of classical poststreptococcal glomerulonephritis has declined in industrialized countries, whereas staphylococcal infection-related glomerulonephritis, often presenting with IgA-dominance on histology, has become increasingly important in the elderly population with comorbidities.

While classical poststreptococcal glomerulonephritis develops after the resolution of infection, most other forms of IRGN, particularly staphylococcal infection-related glomerulonephritis, manifest during ongoing infection.

The pathophysiology of most IRGNs is related to immune complex deposition. Especially in poststreptococcal glomerulonephritis, however, the alternative rather than the classical pathway of complement plays a central role. Poststreptococcal glomerulonephritis and complement 3 glomerulonephritis may therefore show overlapping clinical and histologic characteristics.

IRGNs can mimic other forms of glomerulonephritis (e.g., antineutrophil cytoplasmic antibody-associated glomerulonephritis, IgA nephropathy, and complement component 3 (C3) glomerulonephritis). Therefore a high index of suspicion is required, and the diagnosis of IRGN usually involves a combination of clinical findings, serologic, and microbiological tests and renal biopsy.

There is no specific therapy for IRGN other than treating the underlying infection (if still ongoing) and supportive treatment. Immunosuppressive therapy has a limited role in IRGN and may be harmful if infection has not yet resolved.

Classical poststreptococcal glomerulonephritis, especially in children, has a good prognosis, while other forms of bacterial IRGNs have less favorable outcomes and often lead to persistent chronic kidney disease or even kidney failure.

INTRODUCTION

The association between infections and renal injury has been appreciated since the 19th century.¹⁻⁴ Today, infection-related

glomerulonephritides (IRGNs) are considered among the most frequent subtypes of glomerulonephritis globally. Their epidemiology, however, varies widely by geography. Diverse infectious agents, such as bacteria, viruses, and fungi but also protozoa and helminths, have been implicated in the

Table 34.1 Infectious Organisms Associated with Glomerulonephritis^{29,152,193,217}

Bacterial	Viral
<i>Streptococcus pyogenes</i> , <i>S. viridans</i> , <i>S. mutans</i> , <i>S.</i> <i>zooepidemicus</i> , <i>S. bovis</i> <i>S.</i> <i>pneumoniae</i>	Hepatitis A, B, C, E
<i>Staphylococcus aureus</i> , <i>S.</i> <i>epidermidis</i> , <i>S. hemolyticus</i>	HIV
<i>Enterococcus</i>	Epstein-Barr virus
<i>Listeria monocytogenes</i>	Coxsackie B
<i>Escherichia coli</i>	ECHO virus
<i>Pseudomonas aeruginosa</i>	Cytomegalovirus
<i>Proteus mirabilis</i>	Varicella zoster
<i>Klebsiella pneumoniae</i>	Mumps
<i>Enterobacter cloacae</i>	Rubella
<i>Yersinia enterocolitica</i>	Influenza
<i>Salmonella typhi</i> , <i>S. paratyphi</i> , <i>S. typhimurium</i>	Measles
<i>Campylobacter jejuni</i>	Parvovirus B19
<i>Serratia marcescens</i>	Hantavirus
<i>Brucella abortus</i>	Adenovirus
<i>Legionella pneumophila</i>	Varicella
<i>Acinetobacter baumannii</i>	Yellow fever
<i>Capnocytophaga</i>	Dengue
<i>Bartonella henselae</i>	SARS-CoV2
<i>Coxiella burnetii</i>	
<i>Haemophilus influenzae</i>	
	Fungal
<i>Neisseria meningitidis</i> , <i>N.</i> <i>gonorrhoeae</i> , <i>N. flava</i>	<i>Histoplasma capsulatum</i>
<i>Propionibacterium acnes</i>	<i>Candida</i>
<i>Corynebacterium diphtheriae</i>	<i>Coccidioides immitis</i>
<i>Mycobacterium leprae</i> , <i>M.</i> <i>tuberculosis</i> , <i>M. avium</i>	
<i>Treponema pallidum</i>	
	Helminthic
<i>Mycoplasma pneumoniae</i>	<i>Schistosoma mansoni</i> , <i>S. japonicum</i> , <i>S.</i> <i>haematobium</i>
<i>Chlamydia pneumoniae</i>	<i>Wuchereria bancrofti</i>
<i>Borrelia burgdorferi</i>	<i>Brugia malayi</i>
	<i>Loa loa</i>
	<i>Onchocerca volvulus</i>
Protozoal	
<i>Plasmodium malariae</i> , <i>P.</i> <i>falciparum</i>	<i>Trichinella spiralis</i>
<i>Toxoplasma gondii</i>	<i>Strongyloides stercoralis</i>
<i>Trypanosoma cruzi</i> , <i>T. brucei</i>	<i>Toxocara canis</i>
<i>Leishmania donovani</i>	

development of glomerulonephritis⁵ (Table 34.1). Besides IRGN presenting as acute kidney injury (AKI), it is a potential risk factor for developing chronic kidney disease (CKD) in later life.^{6,7} Unidentified IRGNs may contribute significantly to the prevalence of CKD in resource-constrained regions due to a high prevalence of communicable diseases that can cause glomerulonephritis.⁸ However, given that these conditions are often only diagnosed in retrospect or presumed, their precise contribution to the CKD burden in these areas remains uncertain. Nevertheless, this may explain some of the disparities in end-stage kidney failure (ESKF) burdens globally.

The nomenclature of glomerular diseases in general is evolving and sometimes confusing. This is due to the fact that glomerular diseases can be classified on the basis of their etiology or underlying systemic disease, pathophysiology (e.g., mediated by immune complexes and deposition of monoclonal proteins), histologic pattern, or clinical syndrome they cause.⁹ The same is true for IRGN, where no generally accepted nomenclature exists and multiple terms for partially overlapping disease entities are used in the literature. Definitions may differ between publications, which sometimes renders comparison and interpretation of data difficult. The most common forms of IRGNs in Western societies are bacterial, and some authors use the term IRGN (or infection-associated glomerulonephritis [IAGN]) interchangeably with bacterial IRGN. We will use the term in a wider sense, encompassing glomerulonephritis or glomerular disease caused by any type of infection.

Bacterial IRGN may be further classified on the basis of the causative pathogen, the infection's anatomic location, and histopathologic characteristics of the glomerulonephritis or by the timing of their manifestations in relation to the infection or its resolution (latency) (Fig. 34.1). Historically, poststreptococcal glomerulonephritis (PSGN), which typically occurs after the resolution of an infection of the throat or skin with group A β -hemolytic streptococci, has been the prototypical and most common bacterial IRGN for much of the past century.¹⁰⁻¹² The term postinfectious glomerulonephritis (PIGN) has therefore been used largely synonymous with PSGN (often also called acute poststreptococcal glomerulonephritis [APSGN] because of its acute and usually reversible course). PSGN typically occurs in children¹⁰⁻¹² and remains the most common form of acute glomerulonephritis in this age group, although its incidence has decreased dramatically in economically developed countries.^{13,14} Incidence and prevalence of PSGN strongly depend on the development index of a country.¹⁵ Reasons for the decline in developed countries are improved living conditions, which reduce transmission of the pathogen, and antibiotic treatment of throat or skin infection with antibiotics, which eliminates nephritogenic group A β -hemolytic streptococci from circulating in the community.^{13,15-18} In contrast, the incidence of other forms of bacterial IRGN appears to be rising in adults, particularly in elderly patients in Western societies. These forms of IRGN are often associated with *Staphylococcus* infections, and endocarditis is a frequent site of infection. Histologically, IgA deposits have been recognized as a common feature in such cases. Thus *Staphylococcus*-associated glomerulonephritis (SAGN), endocarditis-associated glomerulonephritis, and IgA-dominant infection-related glomerulonephritis have all been described as distinct entities based on a specific pathogen, a site of infection, or a histologic pattern, respectively (Fig. 34.2). There is considerable overlap between these three entities with many cases fulfilling the criteria for each of them. Nevertheless, these three terms cannot be considered synonymous. SAGN often occurs with sites of infection other than endocarditis and may or may not display IgA deposits on histology. Likewise, IgA-predominant IAGN may be triggered by other pathogens than *Staphylococci*, and endocarditis-associated glomerulonephritis is not limited to *Staphylococci* as causative organisms.¹⁸⁻³² Importantly, these forms of IRGN typically occur with

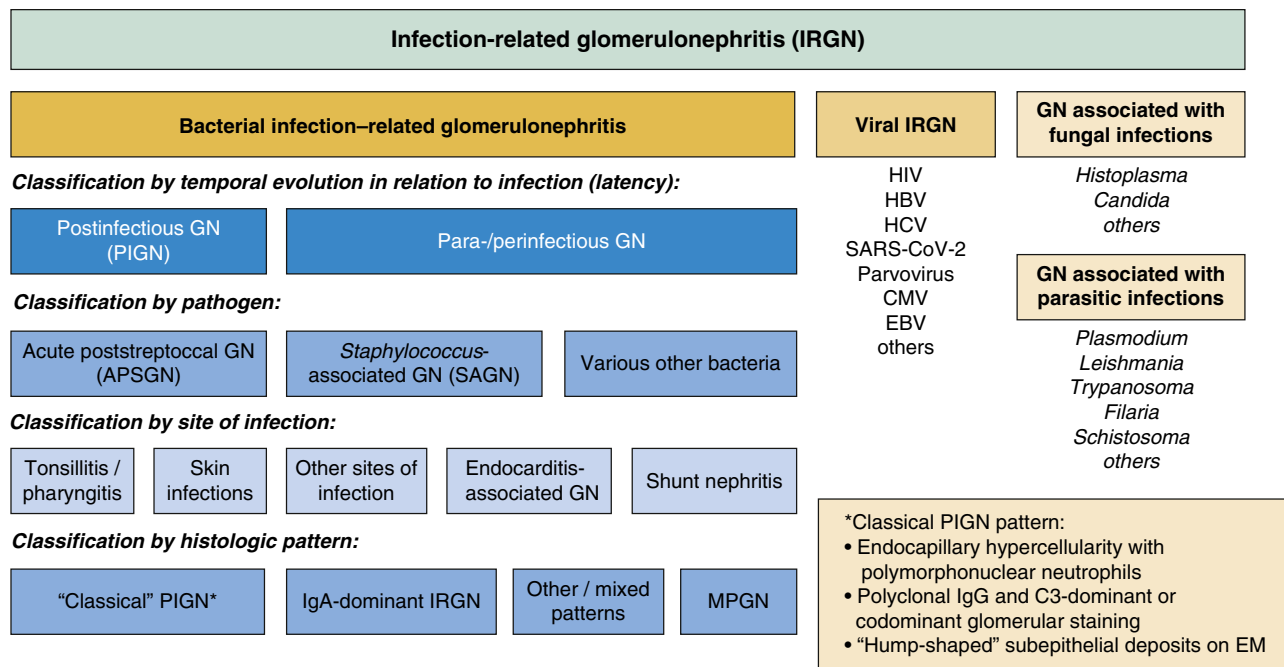


Fig. 34.1 Classification scheme for infection-related glomerulonephritis (IRGN). Bacterial IRGN can be classified according to the temporal association of glomerular disease with infection, the triggering pathogen, site of infection, and histologic pattern. Disease entities have been defined on each of these levels, resulting in various terms that apply to overlapping patient populations. In the scheme shown, pathogens, infection sites, temporal relationships, and histologic patterns often associated with each other are depicted below each other (e.g., acute post-streptococcal glomerulonephritis usually develops with a latent period after a streptococcal infection of the throat or skin and has a characteristic “classical” postinfectious glomerulonephritis pattern on histology).

ongoing infections (i.e., parainfectious or periinfectious rather than postinfectious). This distinction is not purely semantic but of importance for clinical management. In poststreptococcal glomerulonephritis, the infection has resolved and does not require antibiotic treatment. In contrast, treatment of the ongoing underlying infection is crucial in SAGN, whereas steroids and immunosuppressive drugs may be detrimental in these cases.³³

Viral infections most commonly associated with glomerular diseases are hepatitis B and C and HIV.³⁴⁻³⁷ Whereas bacterial IRGNs often display a characteristic histologic pattern that points to bacterial infection as their etiology (although they may be difficult to differentiate from other glomerular diseases like IgA nephropathy or antineutrophil cytoplasmic antibody (ANCA)-associated crescentic glomerulonephritis in some cases), viral infections have been associated with specific glomerular diseases. These diseases may be associated with various etiologies or occur as “primary” glomerular diseases. The causative role of the viral infection has not been doubtlessly established in all cases, and these entities may or may not be included under the umbrella of IRGN. Typical examples include membranous nephropathy associated with hepatitis B virus (HBV) or membranoproliferative glomerulonephritis (MPGN) associated with HCV. A peculiar glomerular pattern, collapsing glomerulopathy (also called “collapsing-variant focal segmental glomerulosclerosis [FSGS]”), has been first described in HIV-positive patients. Histologically, the disease is characterized by dedifferentiation, hypertrophy, and hyperplasia of podocytes, leading to a collapse of underlying glomerular capillaries. Beyond HIV, several other viral infections have been associated with collapsing glomerulopathy, including parvovirus B19, EBV, and most

recently, SARS-CoV-2. However, collapsing glomerulopathy may also be associated with nonviral triggers, such as autoimmune disease or medications.

Chronic parasitic infections such as malaria, schistosomiasis, and filariasis are increasingly recognized as causes of glomerulonephritis, mostly in tropical and subtropical climate zones and especially in socioeconomically deprived regions of the world. Malaria is one of the most prevalent endemic infectious diseases worldwide, affecting approximately 250 million people,³⁸ and malaria-associated glomerulonephritis is estimated to occur in 18% of infected individuals.³⁹ Schistosomiasis affects about 240 million people worldwide,⁴⁰ and histologic studies have demonstrated that glomerular lesions occur in 10% to 12% of patients with schistosomiasis.⁴¹ Unfortunately, established schistosomal glomerulonephritis does not respond to antiparasitic treatment.^{42,43} Roughly, 50 million people are affected by lymphatic filariasis globally,⁴⁴ and studies indicate that approximately 25% of them suffer from glomerular proteinuria.^{45,46} Among individuals with onchocerciasis and loiasis (both filarial disorders), about 3% to 5% develop nephrotic syndrome.^{47,48} These facts indicate that parasite-related glomerulonephritides are a significant and probably underappreciated global cause of kidney disease.

Given the variety of infections associated with glomerular disease, multiple pathogenetic mechanisms have been implicated. The immune system plays a pivotal role in most IRGNs, and the majority of IRGNs are categorized into the histologic and pathogenetic class of immune complex glomerulonephritis (Fig. 34.3).⁴⁹ Particularly within the past couple of years, the involvement of the complement system in many forms of IRGN has been increasingly appreciated. Direct interaction of the pathogen with host cells, however,

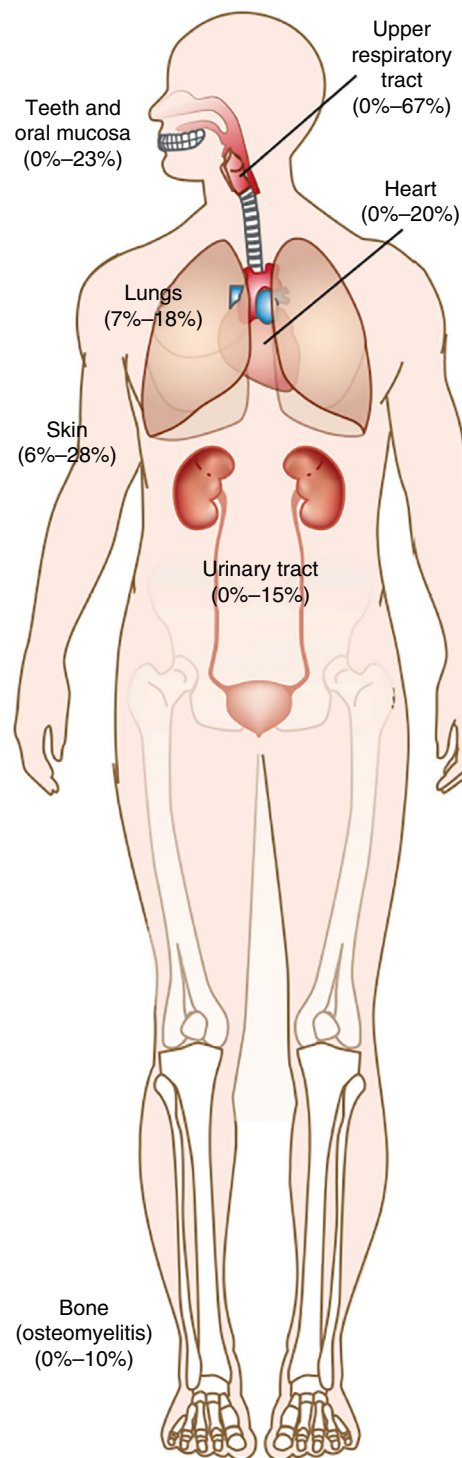


Fig. 34.2 Sites of infection in adult bacterial infection-related glomerulonephritis (IRGN). (With permission from Nasr SH, Radhakrishnan J, D'Agati VD. Bacterial infection-related glomerulonephritis in adults. *Kidney Int.* 2013;83(5):792–803.)

can also cause glomerular injury. In HIV-associated nephropathy (HIVAN), for example, direct viral infection of podocytes and parietal epithelial cells may play an important role and induce dedifferentiation of these cells.⁵⁰ A plethora of histologic lesion patterns can be observed in glomerular diseases associated with infection, and none is specific to IRGN (Table 34.2).

PATHOGENESIS OF BACTERIAL INFECTION-RELATED GLOMERULONEPHRITIS—SPECIAL EMPHASIS ON THE ROLE OF THE COMPLEMENT PATHWAY

PSGN represents the best-studied disease entity within the field of IRGN. While pathophysiology varies with the type of IRGN, certain features of PSGN are shared with other forms of IRGN and may point to more general mechanisms.

IRGN results from a response of the innate and adaptive immune system to a usually extrarenal pathogen (Fig. 34.4). Central mechanisms are immune complex deposition and complement activation mainly via the alternative pathway,^{51,52} resulting in glomerular complement 3 (C3) deposition and reduced serum C3 levels. Bacterial antigens, which can be intracellular, membrane bound, or secreted, probably play a pivotal role. The prevailing theory is that the adaptive immune system mounts an antibody response to these antigens, leading to circulating immune complexes, which are then trapped in the glomeruli within the mesangium or capillary wall. Alternatively, immune complexes can also form in situ from binding of antibodies to either planted bacterial antigens trapped in the glomeruli or host antigens by cross-reactivity.⁵³ Candidate host antigens are collagens, laminin, and heparan sulfate.⁵⁴ Planted bacterial antigens are thought to be mostly cationic because they are prone to pass the negatively charged glomerular wall and deposit in the outer membrane, resulting in the characteristic “hump”-shaped subepithelial deposits.⁵⁵ Immune complexes within the vessel wall then cause the release of cytokines and expression of adhesion molecules, which enable invasion of immune cells. In addition, local production and activation of enzymes like matrix metalloproteinases and plasmin and stimulation of the complement pathway by immune complexes exacerbate glomerular injury.^{15,18}

Major candidate antigens from group A β -hemolytic streptococci, which are potentially nephritogenic in PSGN, are the nephritis-associated plasmin receptor (NAPlr) and streptococcal pyrogenic exotoxin B (SpeB).⁵⁶ However, NAPlr is not specific to group A streptococci and glomerular deposits have been observed in bacterial IRGN elicited by other species of streptococci or different bacteria.^{15,57-62}

A new mechanism of alternative pathway activation in PSGN has been described. Autoantibodies to complement factor B have been detected in the serum of most children with PSGN in the acute phase. These antibodies were found in only few control patients with C3 glomerulopathy (C3G). They were transient and disappeared in most patients during follow-up, correlated inversely with plasma C3 levels, and showed direct correlation with the soluble membrane attack complex C5b-9. Complement factor B antibodies likely enhance the binding of C3b to complement factor B and catalyze the formation of the alternative complement pathway convertase C3bBb, but interestingly, in contrast to C3 nephritic factor (C3NeF, commonly found in C3G), they appear not to *stabilize* the C3 convertase (C3bBb).

The pathogenesis of other forms of bacterial-associated glomerulonephritis, such as *Staphylococcus*-associated glomerulonephritis (SAGN), is less well understood. In SAGN, the continuous deposition of circulating immune complexes during ongoing infection appears to be an important mechanism. These immune complexes can form

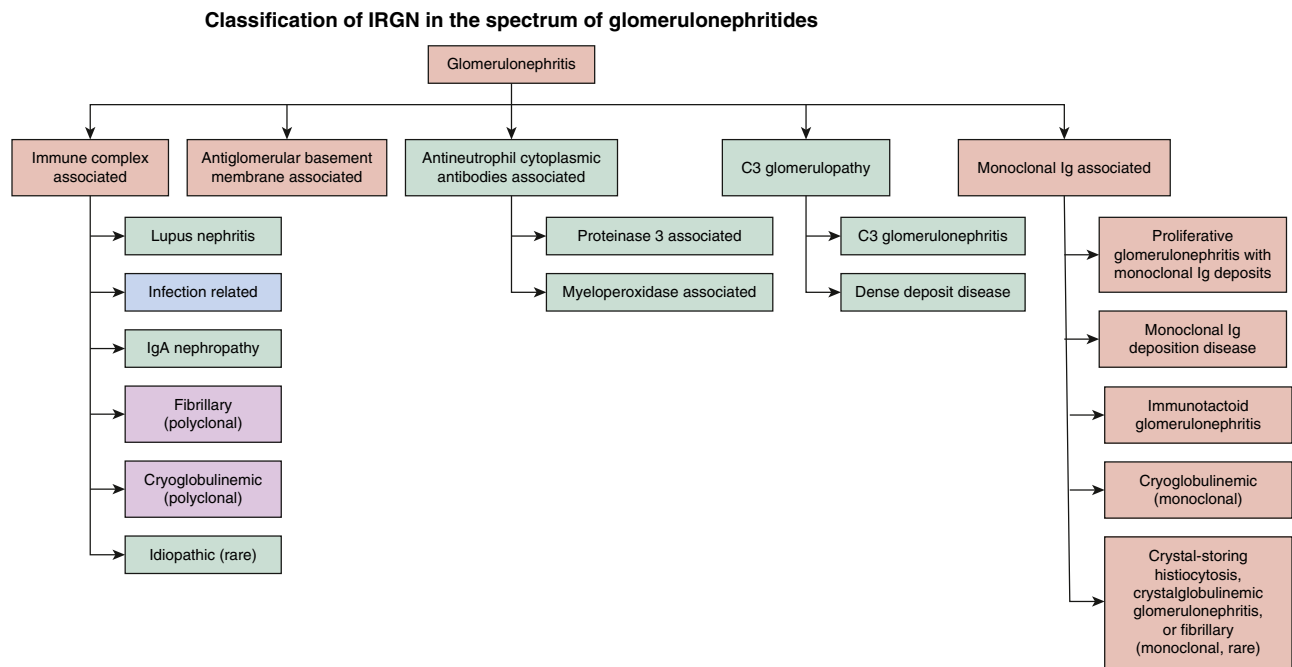


Fig. 34.3 Classification of infection-related glomerulonephritis (IRGN) in the spectrum of glomerulonephritides (GNs) based on pathogenesis (blue).⁴⁹ Violet: specific forms of GN that can be elicited by infections; light green: other forms of GN, which can be mimicked by IRGN.

Table 34.2 Histologic Lesion or Glomerulonephritis Patterns in Infection-Related Glomerulonephritis

Histologic lesions/glomerulonephritis patterns

Endocapillary glomerulonephritis
Exudative glomerulonephritis
Mesangial proliferative glomerulonephritis
Membranoproliferative glomerulonephritis
Necrotizing glomerulonephritis
Crescentic glomerulonephritis
Pauci-immune glomerulonephritis
Collapsing glomerulonephritis
Minimal change glomerulonephritis
Sclerosing glomerulonephritis
Membranous glomerulonephritis

in endocarditis, osteomyelitis, deep abscesses, skin, urinary tract, and pulmonary infections.⁶³ The process requires the constant presence of antigen in the blood.⁶⁴ Impaired clearance of microorganisms due to primary or secondary immunodeficiency is therefore a predisposing condition and explains the increased risk of SAGN in elderly patients and individuals with diabetes and those suffering from alcohol abuse or on immunosuppressive therapies. Other conditions that favor persistence of bacterial antigens and increase the risk for IRGN include prostheses, artificial heart valves, pacemakers, and long-term indwelling catheters.

Bacterial superantigens like staphylococcal enterotoxins might play an additional role in immune complex formation and direct kidney damage. These superantigens are able to massively activate T cells via interaction with MHC class II on antigen-presenting cells, resulting in the release of cytokines

such as TNF α , IL-6, IL-8, and IL-10, which may directly harm renal tissue or lead to polyclonal stimulation of B cells and production of IgA and/or IgG.^{65,66} Why this process only occurs in a minority of patients with staphylococcal infections is unclear but points to an important role of host factors. Overactivation of neutrophils and formation of neutrophil extracellular traps (NETs) has been proposed as another pathomechanism.⁶⁷ The pathogenesis of IRGN caused by gram-negative or other bacteria such as *Mycobacteria* remains largely unexplored, yet also in these entities, immune complex deposition seems to play a major role.

OVERLAP BETWEEN C3 GLOMERULOPATHY AND BACTERIAL IRGN

PSGN and other forms of bacterial-related glomerulonephritis share several features with C3 glomerulopathy (C3G). Both diseases share alternative pathway activation and C3 deposition within the glomeruli as a hallmark, and low plasma C3 frequently occurs in both entities. On renal biopsy, immunoglobulin depositions are often detected in bacterial IRGN but may also be present in C3G. Furthermore, bacterial infections may trigger C3G. These overlaps render distinction of one entity from the other difficult in some cases.⁶⁸⁻⁷¹ Nevertheless, the two conditions have distinct etiologies and clinical contexts. C3G is caused by acquired (autoimmune) or genetic dysregulation of the alternative pathway. In the latter, autoantibodies such as C3, C4, or C5 nephritic factor (C3NeF, C4NeF, and C5NeF) or factor H antibodies lead to overactivation of the alternative pathway. In about 25% of cases, genetic variants in complement genes such as *C3*, *CFB*, *CFH*, *FI*, or *CFHR5* affect the alternative pathway. Uncontrolled and excessive alternative pathway activation results in constant C3 deposition within

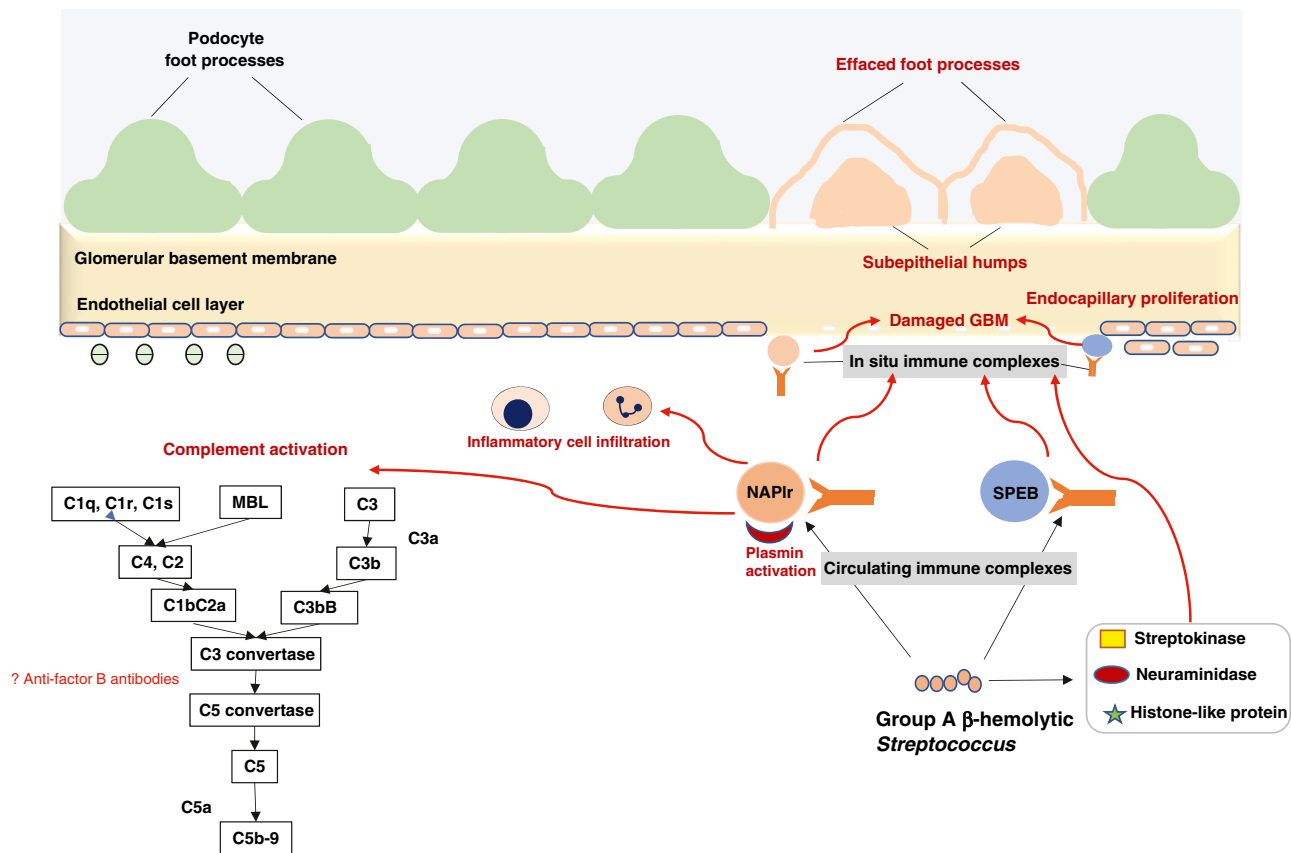


Fig. 34.4 Pathogenesis of poststreptococcal glomerulonephritis (see text for explanations). The image depicts the role of streptococcal antigens in the formation of immune complexes and activation of the complement pathway and resultant changes in the glomerulus.

the glomeruli, inducing damage and inflammation.⁷² In PSGN and other bacterial IRGNs, the alternative pathway is activated by the infection and corresponding immune response. However, in children with classical, self-limiting PSGN, rare complement gene variants can be found with a similar frequency compared with children with C3G and significantly more often than in healthy individuals.⁷³ Likewise, autoantibodies against factor B and C3NeF can be found in both C3G and PSGN, although with somewhat different frequencies. Therefore in cases of PSGN that do not improve on supportive treatment, a diagnosis of C3G must be actively pursued.^{68,69,74,75} Some argue that C3G and C3-dominant IRGN are entities in a single glomerular disease category or disease spectrum.^{60,62,69} Whether complement blockade with newly developed inhibitors of the alternative pathway might be a successful strategy in subgroups of patients with bacterial IRGN will have to be determined in the future.

BACTERIAL INFECTION-RELATED GLOMERULONEPHRITIS

POSTSTREPTOCOCCAL GLOMERULONEPHRITIS

Glomerulonephritis following infection with group A β-hemolytic streptococci is the prototype of IRGN in children.¹² In the early part of the 20th century, the presence of dark and scanty urine was a feared complication during epidemics of scarlet fever.⁷⁶ However, since the mid-1900s,

a notable decline in the prevalence of PSGN was observed, with most cases seen in low-resource regions.¹⁷

The incidence of poststreptococcal glomerulonephritis (PSGN) shows a wide variability ranging from 6 to 24.3 cases per 100,000 person-years in high-income to lower-income countries across the world.¹³

EPIDEMIOLOGY

PSGN is predominantly seen in children; however, it can also affect adults, especially the elderly,^{63,77} those with comorbidities such as diabetes mellitus and malignancy, those on certain drugs (chemotherapeutics or glucocorticoids), or chronic alcoholics. These comorbidities might contribute to an immunocompromised state, which makes individuals more prone to streptococcal infections.^{63,77,78}

PSGN may occur as isolated cases or in clusters during epidemics.^{76,79} Though group A β-hemolytic streptococci account for most cases, epidemics of PSGN with group C streptococcus have also been reported.¹⁵ The nephritogenic strains, identified on the basis of the M protein, cause pharyngitis (strains 1, 2, 4, 12, 18, and 25) and impetigo (strains 49, 55, 57, and 60), which are the most common infections leading to PSGN.^{80,81}

PATHOGENESIS

PSGN is an immune complex-mediated disease. The various mechanisms of glomerular injury in PSGN are described in

Fig. 34.4. The nephritogenic antigens, nephritis-associated plasmin receptor (NAPlr) and streptococcal pyrogenic exotoxin B (SPeB), which are secreted cationic proteins, are key players in glomerular injury.^{76,81} Both proteins have been detected in the glomeruli during the early phase of PSGN, and antibodies to NAPlr and SPeB are present in convalescent sera of PSGN patients.^{15,52} Other important antigens implicated in the pathogenesis are streptokinase, enolase, neuraminidase, histone-like proteins, etc.^{76,81} The immune complexes are either formed by these nephritogenic antigens and antibodies in the circulation or in situ and deposited in the glomerular basement membrane. The immune complexes activate the alternative pathway and induce glomerular inflammation.⁸² Molecular mimicry is implicated in the cross-reactivity of immune complexes formed with the streptococcal M protein against glomerular antigens like laminin and collagen. A component of autoimmunity is also implicated in the pathogenesis, as evidenced by the formation of anti-IgG and factor B antibodies (described in more detail earlier). The modification of IgG by streptococcal neuraminidase and binding of the Fc fragment of IgG to type II receptors on the streptococcal surface are postulated as mechanisms for the production of anti-IgG antibodies.⁷⁶ A genetic susceptibility localized to HLA DR1 and DR4 has also been associated with the development of PSGN.^{82,83}

Streptococci produce streptokinase, which binds to plasminogen to convert it to plasmin.⁶⁰ Plasmin, in turn, increases the invasiveness of the organism. NAPlr released into circulation by streptococci traps plasmin in the glomeruli and causes degradation of the glomerular basement membrane and activation of the complement cascade, and it incites inflammatory cell infiltration within the glomerulus.⁶⁰ Plasmin binds to the endothelium, mesangium, and infiltrating neutrophils.⁶¹ During the early phase of the disease, plasmin deposits have been demonstrated in the glomerulus by direct immunofluorescence.⁶¹ The localization of NAPlr and plasmin to various components of the glomerulus is shown in **Fig. 34.5.**⁷ Plasmin triggers the alternative pathway, releasing anaphylatoxins C3a and C5a, which facilitate the migration of neutrophils and activation of macrophages. In addition, NAPlr can directly activate the alternative pathway via conversion of

C3.⁵⁶ NAPlr also stimulates helper T cells, which in turn activate macrophages. The activated macrophages release growth factors, which cause endothelial and mesangial cell proliferation.^{52,84,85} NAPlr and associated plasmin deposits have been found in other glomerular diseases like ANCA vasculitis, Ig A vasculitis, and MPGN type 1. This entity was previously termed streptococcal infection-related nephritis (SIRN).

Although the alternative pathway is the most important complement pathway in the pathogenesis of PSGN, there is also evidence for activation of the lectin pathway by the streptococcal neuraminidase, and polymorphisms in lectin pathway genes aggravating the immune reactions have been described.^{86,87} The classical pathway plays a minor role and may be activated in the early phase of the disease by the binding of the streptococcal H protein to the Fc receptor of IgG. However, streptococcal chemokine-binding evasins may bind to C4b and prevent activation of the classical pathway.^{52,88} This might be one explanation for the low or absent activity of the classical pathway in PSGN despite immune complex deposition.

Streptococcal pyrogenic exotoxin B (SPeB) was found to colocalize with NAPlr. It has been demonstrated in subepithelial humps⁸⁹ and can prompt the release of cytokines MCP1, IL-1b, TNF, IL-6, IL-8, and TGF- β from leukocytes instigating glomerular inflammation.^{56,90,91} The cationic structure of the toxin increases its affinity to the glomerular basement membrane. Immune complexes formed against SPeB activate plasmin and contribute to the degradation of the glomerular basement membrane. SPeB also plays a role in complement activation.^{83,85} However, SPeB can also block innate host defense via degradation of lytic complement factors.^{52,56,58,87,92}

Several other streptococcal enzymes and molecules are involved in the pathogenesis of glomerular injury. Streptokinase binds and activates plasmin, the complement cascade, and inflammatory pathways. Streptococcal neuraminidase alters IgG by desialylation and results in the production of immune complexes against IgG. Neuraminidase also contributes to the degradation of the glomerular basement membrane and the influx of neutrophils. Streptococcal enolase likewise binds plasmin and contributes to glomerular injury. In addition,

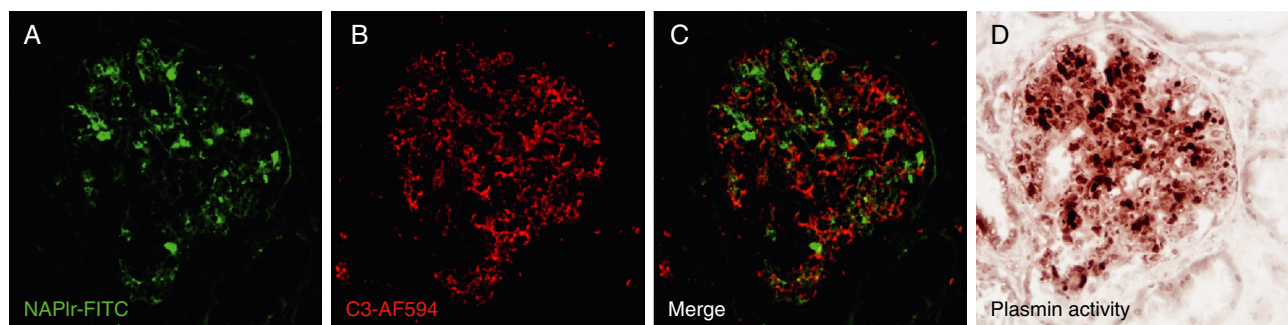


Fig. 34.5 Histology of glomeruli from a patient with poststreptococcal glomerulonephritis. The photomicrographs show histologic staining for complement 3 (C3), nephritis-associated plasmin receptor (NAPlr), and plasmin deposits in the glomeruli of poststreptococcal glomerulonephritis. Though the glomeruli show staining for NAPlr (fluorescein isothiocyanate, green) (A) and C3 (Alexa Fluor 594, red) (B), they are not colocalized (C). Plasmin activity assessed by in situ zymography was found to be positive and had a similar distribution to the NAPlr staining in the glomeruli (D). (Images provided by Dr. Takashi Oda, Professor, Department of Nephrology and Blood Purification, Tokyo Medical University, Tokyo, Japan.)

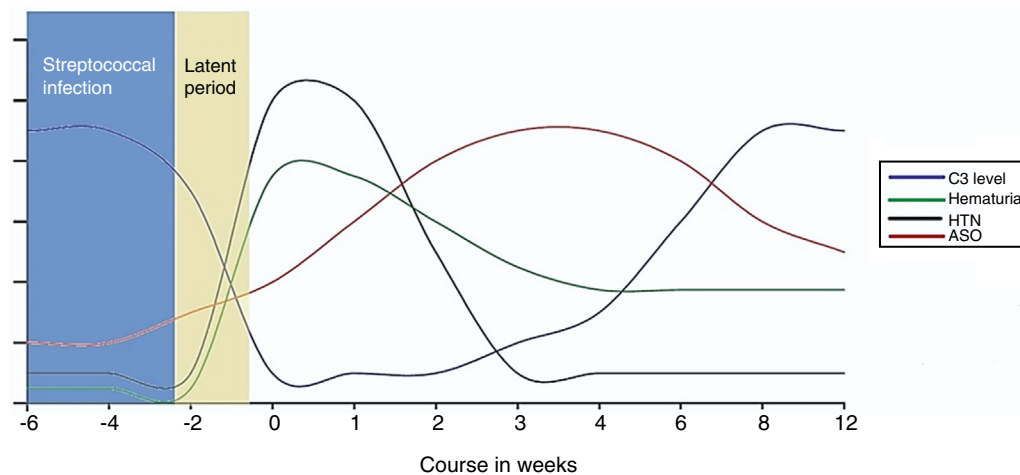


Fig. 34.6 Timeline of the onset of clinical and laboratory features, course of glomerulonephritis, and recovery in relation to the streptococcal infection in poststreptococcal glomerulonephritis. ASO, Antistreptolysin O; C3, complement 3; HTN, hypertension.

histone-like proteins produced by streptococci can activate inflammatory pathways.⁸³

CLINICAL PRESENTATION AND LABORATORY FINDINGS

Streptococcal infections and PSGN are most commonly seen in the age group of 5 to 15 years with a male preponderance.⁸¹ Glomerulonephritis occurs after a latency period of 1 to 2 weeks after streptococcal pharyngitis and 3 to 6 weeks after skin infections.⁹³ The incidence of PSGN is much higher after skin infections (25%) when compared with pharyngitis (5%–10%).^{15,93} The subclinical presentation of PSGN is four to five times more common than overt clinical syndromes. Acute nephritic syndrome is the most common presentation (90%). Less common manifestations are nephrotic syndrome (5%–10%) and rapidly progressive glomerulonephritis (3%–5%).^{76,81,94–97} In children with PSGN, older age, nadir complement C3, and nephrotic-range proteinuria are potential risk factors for kidney injury.⁹⁸ The timeline of the onset, progression, and recovery of PSGN is described in Fig. 34.6.

Laboratory evaluation of PSGN includes urinalysis, which demonstrates subnephrotic-range proteinuria, dysmorphic red blood cells, and red blood cell casts. Complement C3 levels are low in more than 95% of children with PSGN.⁹⁶ Complement C3 levels decline early in the course of the disease during the latent phase and return to normal in 6 to 8 weeks.^{18,96} Clinical evidence of a recent streptococcal infection may not be found in all patients. An antistreptolysin O titer >200 international units or >330 Todd units or a rising titer of antistreptolysin O or anti-DNAse B provides serologic evidence of previous streptococcal infection but is not diagnostic of streptococcal infection.⁹⁶ The antistreptolysin O titers peak at 3 to 5 weeks and anti-DNAse B at 6 to 8 weeks after infection.⁹⁶ The sensitivity and specificity for the combination of antistreptolysin O and anti-DNAse B are 96% and 89%, respectively.⁹⁹ The sensitivity and specificity of anti-DNAse B may be higher when compared with antistreptolysin O for PSGN after skin infections.⁹⁴

Although a kidney biopsy is not required to establish the diagnosis of PSGN in children, it may be indicated when

there is no evidence of infection, in cases with rapid and progressive deterioration in kidney function, prolonged and atypical clinical course, or associated comorbid conditions that confound the diagnosis. Diagnosis of PSGN in adults may be more difficult because the comorbid immunocompromised state and comorbidities can mask symptoms of the infection. Also, clinical signs associated with glomerulonephritis (edema, hypertension, and renal dysfunction) might be attributed to comorbidities, such as diabetes, heart failure, or preexisting arterial hypertension. Since PSGN is a much less frequent cause of acute glomerulonephritis in adults, we suggest renal biopsy even in patients with a typical history to confirm diagnosis. PSGN in the elderly tends to have a more severe and prolonged course, often with incomplete recovery.⁶³

An alternate diagnosis should be considered when there are systemic symptoms, persistently elevated or increasing creatinine, and persistent hypocomplementemia.

HISTOPATHOLOGY

The histopathologic hallmark of PSGN is diffuse, exudative proliferative glomerulonephritis with immunofluorescence showing capillary and mesangial staining for IgG and C3 (Fig. 34.7). Crescents are rarely seen in PSGN. In the late phase of illness, the biopsy may only show mesangial hypercellularity with lymphocytic infiltration.⁸¹ The findings on immunofluorescence also change from deposits of C3 and IgG in the capillary walls and mesangium (starry sky appearance) in the early phase to resorption of capillary wall deposits (mesangial pattern) in later phases. Coarse granular deposits along the capillary wall create a “garland” pattern in some cases.¹² Electron microscopy demonstrates the characteristic subepithelial humps (Fig. 34.8). Some deposits may be seen in the mesangial and subendothelial areas as well.¹²

MANAGEMENT

PSGN is most often a self-limiting condition and requires supportive treatment only. Resolution of clinical symptoms occurs within 2 to 3 weeks. The levels of complement C3

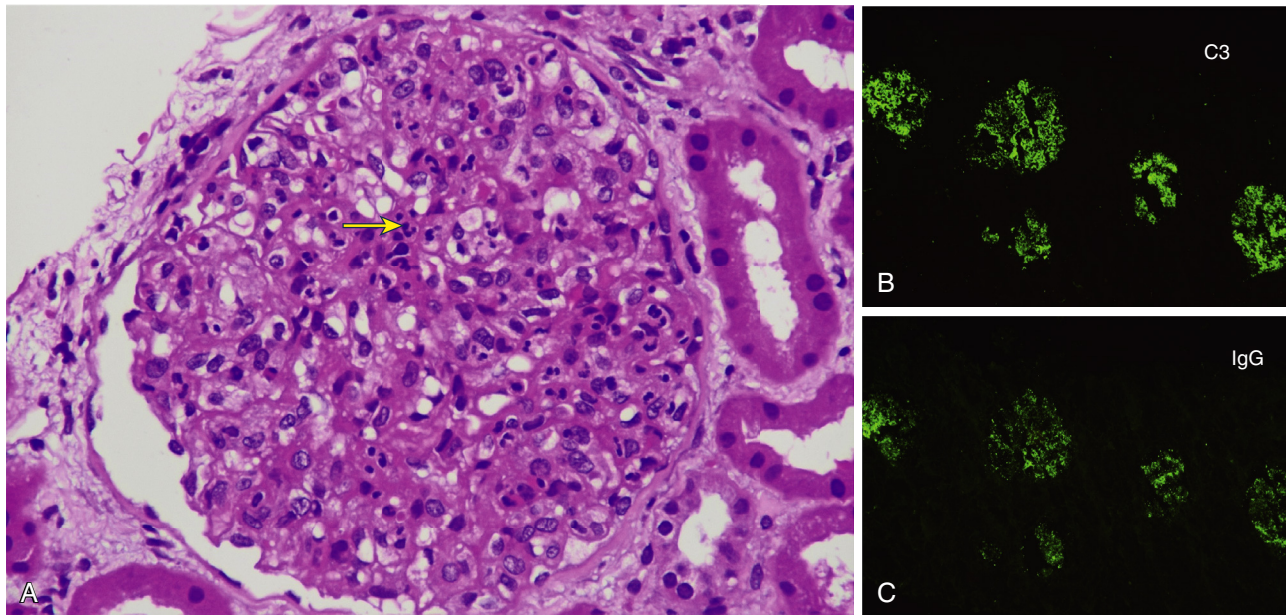


Fig. 34.7 Histology of glomeruli from a patient with poststreptococcal glomerulonephritis. The light microscopic picture (hematoxylin-eosin stain) (A) shows endocapillary and mesangial proliferation with neutrophilic infiltration (*yellow arrow*). The immunofluorescence shows granular deposits of C3 (B) and IgG (C) deposits along the glomerular capillary loops and mesangium. C3, Complement 3; IgG, immunoglobulin G. (Images kindly provided by Dr. Anila Abraham, Nephropathologist, Renopath, Chennai, India.)

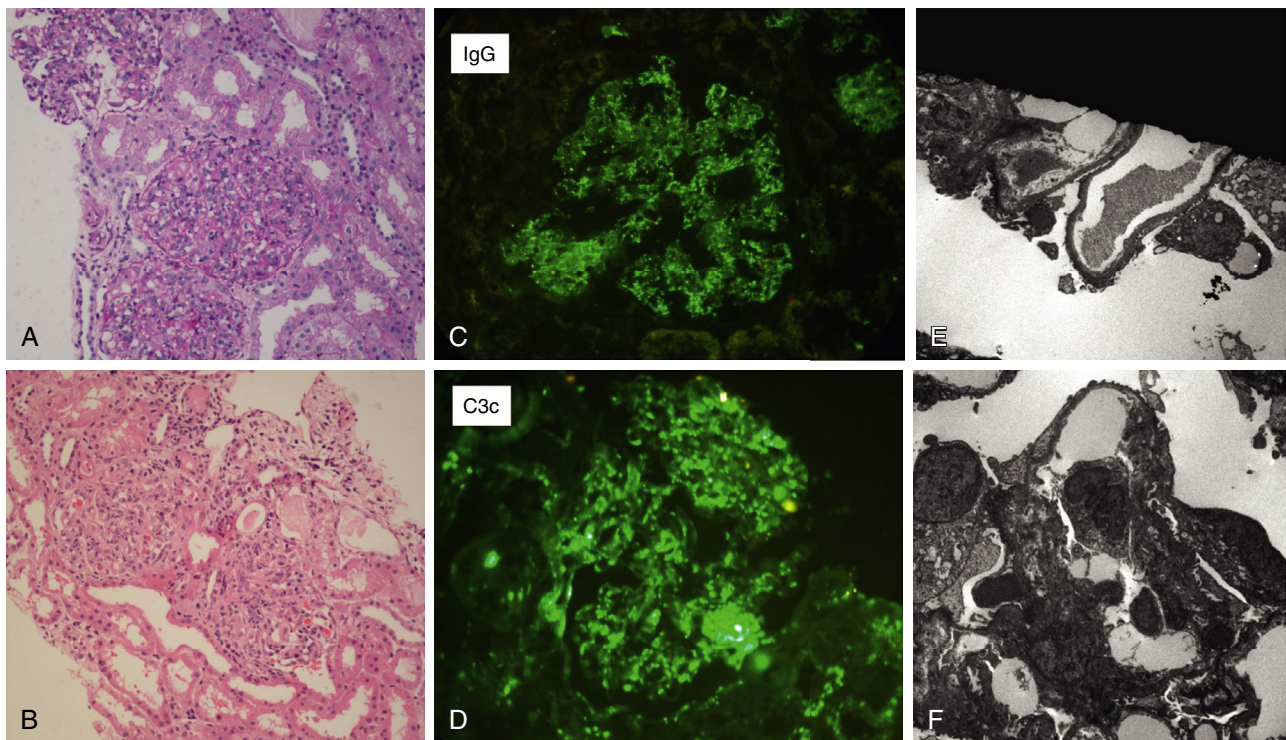


Fig. 34.8 Histology of glomeruli from a patient with poststreptococcal glomerulonephritis. (A and B) Periodic acid–Schiff and hematoxylin-eosin–stained sections, respectively, showing a proliferative, exudative pattern of injury. (C and D) From direct IF study showing peripheral and mesangial coarse granular deposits of IgG and C3c with C3c deposits being dominant. (E and F) From electron microscopy showing subepithelial humps. (Histopathology images have been provided by Dr. Swarnalatha Gowrishanker, Nephropathologist, Apollo Hospital, Hyderabad, India.)

return to normal after 6 to 8 weeks. Microscopic hematuria, however, may persist for 1 to 2 years after an episode of PSGN.^{94,100} Recurrence of PSGN is extremely rare.^{94,96}

Furosemide is the drug of choice in children with features of hypervolemia and hypertension. A calcium channel

blocker may be added to control hypertension.^{18,96,101} Angiotensin-converting enzyme inhibitors are generally avoided in the acute phase of the illness because they may reduce the glomerular filtration rate and worsen hyperkalemia. Notably, a large retrospective observational

study from India suggests a better outcome with renin-angiotensin system (RAS) blockade in adult patients with bacterial IRGN; however, patients fulfilling the criteria for PSGN were in the minority in the population investigated.¹⁰² Restriction of salt, fluid, and potassium intake is recommended in the early phase of the disease. Intravenous antihypertensive medications may be required in children presenting with acute severe hypertension.⁹⁶ Dialysis is rarely needed, except in some children presenting with rapidly progressive glomerulonephritis.⁹⁶

The role of immunosuppressants even in the case of crescentic glomerulonephritis is limited.¹⁰³ Immunosuppression with corticosteroids and alkylating agents has been used in cases with crescentic glomerulonephritis,¹⁰⁴ but there is little evidence of benefit on long-term outcomes.¹⁰⁵ Even among adults, the role of steroids in the treatment of PIGN is restricted to severe cases, although the benefits seem to be limited.¹⁰⁵⁻¹⁰⁷

Antibiotic therapy is not warranted in all cases of PSGN because the infection has subsided by the time the nephritis develops. Antibiotic therapy does not reduce the risk of developing PSGN.^{103,108} Antibiotic prophylaxis with penicillin may be advised for close contact with children with streptococcal infection, during epidemics, and in those with a streptococcal carrier state.¹⁰¹ Two vaccines are under development for PSGN. The MJ8VAX consisting of a 29 amino acid peptide of the carboxy terminal of M protein as antigen is in phase 1 trials. The StrepAnova, a 30-valent vaccine with antigen derived from M protein, is in phase 2 trials.¹⁰⁹

The most common differential diagnoses are C3 glomerulopathy (see earlier), systemic lupus erythematosus, and IgA nephropathy.^{76,100}

The prognosis of PSGN is good, and complete recovery is seen in most children.⁹⁷ Long-term follow-up studies in children found that about 20% of children with PSGN had persistently abnormal urinalysis; however, less than 1% developed kidney failure.¹¹⁰ The prevalence of proteinuria (7.2%), microscopic hematuria (5.4%), arterial hypertension (3%), and azotemia (0.9%) on follow-up was comparable with that in the normal population.⁷⁶ Children from low-resource countries with more severe disease at onset or AKI requiring dialysis had a higher incidence of sequelae including proteinuria, hypertension, and azotemia.¹⁵ The risk of proteinuria and chronic kidney disease (CKD) in young adults has been associated with a combined history of low birth weight and PSGN, especially in those with subsequent obesity.¹¹¹ Outcomes in adults with PSGN are worse than those in children, with residual hypertension, glomerulosclerosis, or CKD being seen in 30% to 50% of elderly patients,¹¹² although a remission rate of 83% has been reported in more current and younger cohorts.¹⁰²

STAPHYLOCOCCUS INFECTION-ASSOCIATED GLOMERULONEPHRITIS (SAGN) AND IGA-DOMINANT INFECTION-RELATED GLOMERULONEPHRITIS (IGA-IRGN)

EPIDEMIOLOGY

Compared with PSGN, *Staphylococcus*-associated glomerulonephritis (SAGN) occurs at the time of infection and is therefore not considered a postinfectious glomerulonephritis.³³

The true incidence of SAGN is unknown, but it is assumed to be the most frequent single cause of IRGN in adults today. Retrospective analyses indicate that approximately 0.1% to 0.9% of all kidney biopsies exhibit SAGN.^{24-28,30,63} SAGN can be caused by *Staphylococcus aureus* (methicillin-resistant/non-methicillin-resistant strains), *Staphylococcus epidermidis*, or *Staphylococcus hemolyticus*.²⁹ *S. epidermidis* is a constituent of the normal skin flora and can survive on biofilms on indwelling catheters or medical devices such as pacemakers.¹¹³ The increasing number of hospitalized elderly and/or immunocompromised patients with indwelling catheters or implantable devices in past decades may have contributed to the increasing incidence of SAGN.

CLINICAL PRESENTATION AND LABORATORY FINDINGS

Patients typically present with AKI, hematuria, and glomerular proteinuria, often in the nephrotic range. Serum C3 may be normal or low.^{30,63} Sometimes, the underlying staphylococcal infection is obvious, such as in superficial skin infections or dental abscesses. Other infection sites, in contrast, may not be obvious, as in the case of chronic osteomyelitis, endocarditis, prosthesis, or pacemaker wire infections. About 20% of patients with SAGN may develop concomitant leukocytoclastic vasculitis.¹¹⁴ Moreover, some patients with SAGN, particularly in the setting of endocarditis, are ANCA positive.^{30,32} Thus SAGN may mimic IgA vasculitis or ANCA-associated vasculitis, and a high index of suspicion is required.

HISTOPATHOLOGY

SAGN shows histologic characteristics of an active proliferative immune complex glomerulonephritis.¹⁸ Light microscopy can range from isolated mesangial proliferation to diffuse endocapillary proliferative lesions and may show segmental glomerular capillary necrosis and cellular crescents in approximately 30%–40% of patients.^{30,63} FSGS lesions are infrequent due to the rather acute nature of the glomerulonephritis but can reflect underlying comorbidities such as obesity or arterial hypertension. Immunofluorescence typically reveals C3- and IgA-positive immune complex deposits in the mesangium and, to a lesser extent, along the glomerular capillary loops. Due to the characteristic IgA positivity, IgA-dominant IRGN (IgA-IRGN) is often used synonymously with SAGN. However, trace or absent IgA staining does not exclude SAGN.²⁴⁻²⁸ Vice versa, IgA-IRGN has been observed in association with pathogens other than *Staphylococcus*.³¹ Importantly, there are no pathognomonic features of SAGN and the histologic picture may vary. C3 dominance or codominance with respect to IgA usually distinguishes SAGN from IgA nephritis but may be mild or negative in some cases, rendering this differential diagnosis is sometimes challenging. In rare cases, both IgA and C3 staining may be minimal or absent, resulting in a pauci-immune picture, such that SAGN may resemble ANCA-associated vasculitis, especially in the presence of ANCA positivity.^{26,30} On electron microscopy, electron-dense deposits are usually observed in the mesangium but may also be detected in the subendothelial or subepithelial space. Large subepithelial humps, much less common than in PSGN, occurred in approximately 30% of the cases in one series.³⁰

MANAGEMENT

In addition to supportive care, treatment should aim at eliminating the causative infection by targeted antibiotic therapy of adequate duration and, if necessary, surgical intervention. Because infection may not be clinically evident in all cases, an infectious focus must be actively pursued by all means (wound swabs, blood cultures, dental radiograph, transthoracic and/or transesophageal echocardiography, computed tomography, magnetic resonance imaging, or positron emission tomography–computed tomography imaging) if SAGN is suspected. In patients with stable kidney function, RAS blockade might be beneficial.¹⁰² Immunosuppressive treatment should not be used during ongoing infection because there is no good evidence of benefit, while the ongoing infection might worsen and additional harm may be caused.^{29,63,65,102,115-119} In a single-center, open-label randomized controlled trial in 52 patients with acute bacterial IRGN in India (not restricted to SAGN), treatment with high-dose glucocorticoids after appropriate antibiotic treatment did not improve renal recovery after 6 months but resulted in a sixfold increase in adverse events.¹⁰⁷

The outcome of SAGN is rather poor. A potential reason is that the often-affected elderly and comorbid patients frequently have preexisting kidney damage. Furthermore, the typically subacute or chronic nature of the infection might lead to accruing damage over time. In addition, antibiotic treatment might take a prolonged time to eliminate the infection in cases of osteomyelitis, inoperable abscesses, or prostheses, which cannot be removed. Most retrospective series in the literature do not present isolated outcomes in SAGN but report combined outcomes of mixed cohorts with bacterial infection–related or “postinfectious” glomerulonephritis. Extrapolation from these data shows that about 40% of patients progress to kidney failure and approximately 40% to 80% suffer from persistent kidney injury.^{29,31,63,65} Risk factors for negative outcomes are preexisting CKD, age, diabetes mellitus, diabetic glomerulosclerosis, and significant interstitial fibrosis/tubular atrophy on kidney biopsy.^{18,31,63,120}

INFECTIVE ENDOCARDITIS-ASSOCIATED GLOMERULONEPHRITIS

EPIDEMIOLOGY

Glomerulonephritis is a well-known complication of infectious endocarditis and was described more than 100 years ago.¹²¹ Twenty percent or more of patients with infective endocarditis may develop glomerular disease.¹²² The pathogen spectrum of infective endocarditis has changed over the past decades, with *S. aureus* becoming the most important pathogen. Likewise, in one of the largest and best-documented recent studies investigating infective endocarditis–related glomerulonephritis, the most common causing microorganism was *S. aureus*.³² As outlined in the introduction, infective endocarditis–related glomerulonephritis therefore shows considerable overlap with SAGN. The second most common pathogen associated with infective endocarditis–related glomerulonephritis is *Streptococcus* sp. and, less frequently, *Bartonella henselae*, *Coxiella burnetii*, *Cardiobacterium hominis*, and *Gemella* were causative.³² However, many other microorganisms might

cause infective endocarditis–related glomerulonephritis, and in some cases, no pathogen can be cultured.^{29,32} Frequent risk factors and comorbidities include cardiac valve disease, intravenous drug abuse, hepatitis C, and diabetes mellitus.

CLINICAL PRESENTATION AND LABORATORY FINDINGS

AKI with hematuria is the most common presentation. Only a minority of patients exhibit typical features of acute nephritic syndrome or rapidly progressive glomerulonephritis.³² Therefore infective endocarditis–related glomerulonephritis must be considered in the differential diagnosis of AKI in patients with endocarditis and discriminated from other causes of AKI in this setting, such as embolic infarctions, drug-induced interstitial nephritis, drug toxicity, or hemodynamic AKI.

Infective endocarditis–related glomerulonephritis may be associated with systemic vasculitis.¹²³ Some patients can develop alveolar hemorrhage, which may lead to a diagnostic dilemma because this presentation imitates antiglomerular basement membrane disease or ANCA-associated vasculitis.¹²⁴ Infective endocarditis may also mimic IgA vasculitis by presenting with purpura and glomerulonephritis with dominant or codominant IgA staining on renal biopsy.¹²⁵ Patients with infective endocarditis, especially associated with *Bartonella* but also with other microorganisms, can develop ANCA.^{29,126} Mostly, these are PR3-ANCAs, but MPO-ANCAs have also been described, and some patients are double positive. The pathophysiologic relevance of ANCAs in renal impairment in this situation is unclear.^{32,127-130} ANCAs may cause a diagnostic dilemma in this setting. Low serum C3 levels and immune deposits on renal biopsy can be a hint pointing at an infectious etiology rather than genuine ANCA-associated vasculitis. Up to 60% of patients with infective endocarditis–associated glomerulonephritis have low C3.¹⁸ Libman-Sacks endocarditis in systemic lupus erythematosus (SLE) with lupus nephritis represents another differential diagnosis from infective endocarditis–related glomerulonephritis. Serologic detection of anti-DNS antibodies and “full-house” staining on renal biopsy usually enable the correct diagnosis.¹³¹

HISTOPATHOLOGY

Diffuse or focal crescentic glomerulonephritis with or without necrosis was the most common (53%) histologic pattern in infective endocarditis–related glomerulonephritis in the largest series, followed by focal or diffuse proliferative glomerulonephritis (37%) and mesangial hypercellularity (10%). C3 staining was most commonly present (94%) and in 37% C3, *only* staining was observed. IgA-dominance or codominance with IgG was present in 17%. Forty-four percent of biopsies were considered pauci-immune on immunofluorescence. Only 14% of patients had humps on electron microscopy. Most deposits on electron microscopy were mesangial (84%), followed by subendothelial (45%), and 10% had no deposits.³² Another study reported that up to two thirds of patients had a pauci-immune crescentic pattern.¹¹⁹ In patients with subacute endocarditis, a membranoproliferative pattern may prevail, reflecting a more protracted glomerular injury.¹⁸

MANAGEMENT

The most important therapeutic measure in infective endocarditis-associated glomerulonephritis is antimicrobial therapy, and surgical valve replacement if necessary and possible to promptly eradicate the source of infection. Yet prognosis is often poor despite these measures. In the study mentioned earlier, 21% of patients with infectious endocarditis-associated glomerulonephritis died. Of those who survived, only 32% had complete recovery of kidney function, 37% had persistent kidney impairment, and 10% progressed to renal failure.³² Other studies report renal failure in up to 50% of cases.^{132,133}

No controlled studies have investigated immunosuppressive treatment in infective endocarditis-related glomerulonephritis. Glucocorticoids alone or in combination with cyclophosphamide have been used^{32,134,135} but are of unproven benefit and carry a substantial risk.

SHUNT NEPHRITIS

Shunt nephritis was first reported in 1965 and arises from infected ventricular shunts implanted for hydrocephalus.^{136,137} It usually develops within 5 years after implantation but may occur much later.¹³⁸ Patients often show systemic signs and symptoms of ongoing inflammation, such as recurring fever, hepatosplenomegaly, and leukocytoclastic vasculitis of the skin. Most patients have hematuria and subnephrotic or nephrotic-range proteinuria. Kidney function may be normal but can range all the way to rapidly progressive glomerulonephritis.^{139,140} The most common causative organism is *S. epidermidis*, but *Propionibacterium acnes* may also occur. Both might be mistaken as a contaminant. Other pathogens associated with shunt nephritis include *S. aureus*, *Escherichia coli*, *Corynebacterium* sp., *Pseudomonas* sp., and fungi such as *Fusarium*.^{140,141} Cerebrospinal fluid and blood cultures are often negative (in 27% and 57%, respectively).¹³⁸ Serologic workup often reveals low C3, and ANA, ANCA, or rheumatoid factor may be positive.^{29,142} Typically, renal biopsy displays a membranoproliferative pattern, with immunofluorescence depicting granular subendothelial and mesangial deposits, often containing polyclonal immunoglobulins (IgM and IgG) and C3. Crescents may be observed.¹⁴³ On electron microscopy, deposits are subendothelial and mesangial.²⁹ Therapy should include immediate surgical removal of the shunt along with antibiotic treatment.¹⁴⁴ Renal outcome is good if diagnosis is made early and therapy is instituted timely.¹⁴⁰ However, delayed diagnosis and late shunt removal are associated with a poor renal and neurologic prognosis even with pathogen-directed antibiotic treatment.^{138,142} There is no established role for immunosuppression in shunt nephritis, and its use should be avoided.

Since ventriculoperitoneal (VP) shunts have largely replaced ventriculoatrial (VA) shunts, the incidence of shunt nephritis has dramatically decreased.^{138,142,145} However, in individuals with VA or VP shunts who present with nephritic or nephrotic syndrome and proliferative glomerulonephritis on renal biopsy, the index of suspicion must still be high.

OTHER FORMS OF BACTERIAL INFECTION-RELATED GLOMERULONEPHRITIS AND THE ROLE OF LATENCY

Apart from the entities described earlier that are defined on the basis of a specific pathogen, a site of infection, or a

histologic characteristic, many cases of glomerulonephritis associated with bacterial infections do not fit into any of these categories. In a retrospective study from India of adult patients with bacterial IRGN, which represents the largest reported cohort of bacterial IRGN to date, a considerable number of patients had gram-negative infections and the urinary tract was the most frequent site of infection.¹⁰² This study also analyzed the association of latency from time of infection to onset of the glomerular disease with pathogens, as well as its role regarding outcomes. Latency was defined as “parainfectious” if glomerular disease was diagnosed with ongoing infection, “periinfectious” if features of glomerular disease ensued within 1 to 7 days of the resolution of infection, or “postinfectious” when at least 7 days had passed between resolution of infection and onset of glomerular disease manifestations. In this cohort, the most common causative agents in postinfectious glomerulonephritis were Streptococci, and the most common site of infection was the skin. Parainfectious glomerulonephritis was commonly associated with gram-negative bacteria and urinary tract infections and thus mostly did not fit into any of the previously described entities (i.e., SAGN or infective endocarditis-related glomerulonephritis). Renal outcomes were worst in cases of parainfectious glomerulonephritis. RAS blockade was associated with better kidney survival, whereas glucocorticoid treatment had no effect, similar to previous studies.^{63,119,146-149}

VIRAL INFECTION-RELATED GLOMERULONEPHRITIS

Glomerulonephritis associated with various viruses has been reported in adults and children as outlined in [Table 34.1](#).¹⁵⁰⁻¹⁵²

HEPATITIS-ASSOCIATED GLOMERULONEPHRITIS

The most common forms of viral IRGNs are due to infection with hepatitis B and C. Hepatitis E virus may also cause glomerulonephritis, especially in chronic infections in immunocompromised hosts.¹⁵³⁻¹⁵⁵

HEPATITIS B VIRUS (HBV)-ASSOCIATED GLOMERULONEPHRITIS

Approximately 240 to 300 million people worldwide are living with chronic hepatitis B,^{156,157} and 3% to 5% of them may develop kidney disease.¹⁵⁸ The incidence of HBV-associated glomerulonephritis in Europe and the United States is lower than in Asia and Africa due to a lower prevalence of chronic HBV infections.¹⁵⁹ HBV immunization can successfully prevent infection and consequently development of HBV-associated glomerulonephritis in patients who have not been infected. The most common form of HBV-associated glomerulonephritis is membranous nephropathy. However, HBV may also cause membranoproliferative glomerulonephritis in the setting of virus-associated mixed cryoglobulinemia.^{36,160-162} Interestingly, patients with HBV-associated membranous nephropathy more often have lower complement levels compared with patients with non-infection-associated forms of membranous

nephropathy.¹⁶³ PLAR2 antibodies can be positive in HBV-associated membranous nephropathy, even if less frequent compared with primary membranous nephropathy.^{164,165} HBV-associated membranous nephropathy may resolve spontaneously in children, especially if seroconversion from HBeAg to anti-HBe occurs.¹⁶⁶ Spontaneous resolution is less frequent in adults.^{35,167} Therapy consists of treating the underlying infection with interferon- α or antiviral medication. Lamivudine is not recommended due to the high resistance rates. Entecavir or tenofovir alafenamide fumarate (TAF) are preferred for first-line therapy.^{160,168}

HEPATITIS C VIRUS–ASSOCIATED GLOMERULONEPHRITIS

Glomerulonephritis occurs in up to half of patients with HCV infection.¹⁶⁹ The most common form is MPGN in the setting of type 2 cryoglobulinemia. Less frequently, HCV-positive patients may develop membranous nephropathy; however, the causal role of the virus is debated.^{170–174} Other, less common forms of glomerulonephritis that have been described in HCV-positive patients in the literature are FSGS, IgA nephropathy, fibrillary, and immunotactoid, but causality has not been proven.¹⁷⁵ Novel treatments for HCV have largely eliminated HCV-related glomerulonephritis.¹⁷⁶ Patients with HCV-associated glomerulonephritis should be treated with direct-acting antiviral therapy. The choice of regimen should be based on GFR, drug-drug interactions, previous treatment, stage of liver fibrosis, and comorbidities. If pangenotypic regimens are not used, HCV genotype should guide therapy in combination with the features mentioned earlier.¹⁷⁷

In patients with moderate to severe cryoglobulinemia and glomerulonephritis (see Chapter 35), immunosuppressive drugs such as systemic glucocorticoids and rituximab are recommended, in addition to antiviral therapy. In the most severe and life-threatening cases, plasmapheresis (PLEX) can be added. PLEX should also be considered before rituximab infusion in patients with a high cryocrit (>10%) to avoid severe systemic reactions due to direct interaction of rituximab with cryoglobulins, which can lead to serum sickness or vasculitis.^{178,179} Nephrologists dealing with HCV-associated glomerulonephritis need to collaborate closely with infectious disease specialists and/or hepatologists to optimize treatment regimens and manage complications.

HIV-ASSOCIATED GLOMERULONEPHRITIS

HIV can cause different forms of glomerulonephritis,⁵⁰ the most common being HIVAN and HIV-associated immune complex kidney disease (abbreviated HIVICK or HIVICD).

HIVAN

Histologically, HIVAN manifests as a collapsing FSGS with concomitant tubular microcysts, tubulointerstitial inflammation, and tubuloreticular inclusion bodies in glomerular endothelial cells on electron microscopy (see Africa, Chapter 75). It classically presents with proteinuria in the nephrotic range and decreased kidney function. Patients are usually not hypertensive and display normal-sized or enlarged kidneys. As stated earlier, glomerulonephritis

ensues from direct damage of visceral and parietal epithelial cells by the virus, which also affects tubular epithelial cells.^{37,50} Classical HIVAN occurs almost exclusively in individuals of African descent and is more frequent in patients from sub-Saharan Africa, where approximately three quarters of the globally affected 40 million patients with HIV infection live.¹⁸⁰ This is largely due to the higher prevalence of apolipoprotein L1 (APOL1) risk alleles, which also confer a higher risk for CKD due to non-HIV diseases, such as FSGS or hypertensive nephrosclerosis.^{181,182} The association of high-risk APOL1 genotypes and collapsing glomerulonephritis has also been described in glomerulonephritis associated with other viruses, such as cytomegalovirus, parvovirus B19, and EBV.^{183–185}

Similar to HCV, highly active antiretroviral therapy has greatly reduced the incidence of HIVAN in patients undergoing therapy and is the treatment of choice for affected individuals. RAS blockade might be helpful to reduce proteinuria.^{37,186,187} In patients with severe interstitial inflammation, glucocorticoids might be beneficial but should be used cautiously.¹⁸⁸ Recently, it has been shown that a small molecule inhibitor (inaxaplin) of APOL1 function can reduce proteinuria in patients with primary FSGS homozygous for the APOL1 risk alleles G1 or G2.¹⁸⁹ Whether this approach will also benefit patients with HIVAN needs to be determined.

HIV-ASSOCIATED IMMUNE COMPLEX KIDNEY DISEASE

Contrary to HIVAN, HIVICK is caused by the deposition of immune complexes and is therefore a primarily immune-mediated kidney disease. Immune complexes consist of viral antigens, antibodies against the virus, and other circulating immune proteins. Once deposited, the immune complexes trigger the complement system, leading to inflammation and activation of the immune response within the glomeruli. In relation to HIVAN, HIVICK seems to have increased in prevalence and become the most common glomerulonephritis form in HIV patients.¹⁹⁰ The specific pattern can vary among individuals. Subtypes of HIVICK are diverse and include MPGN, IgA nephropathy, lupus-like glomerulonephritis, membranous nephropathy, and immunotactoid, fibrillary, and cryoglobulinemic glomerulonephritis. Effective management of HIVICK involves a multifaceted approach. It typically requires antiretroviral therapy to suppress viral replication and reduce immune complex formation. RAS blockade might be of additional benefit. It is also important to treat comorbid diseases, which can contribute to diseases such as hepatitis C or syphilis. The role of immunosuppressive therapy is controversial.¹⁹⁰

Overall, the management of HIV-associated glomerulonephritis requires a multidisciplinary effort involving nephrologists, infectious disease specialists, and other specialists, highlighting the intricate connection between the immune system, viral infections, and renal pathophysiology in this unique subset of glomerulonephritis. More specific aspects of HIV-associated kidney diseases are discussed elsewhere in this book.

COVID-19–ASSOCIATED GLOMERULONEPHRITIS

The COVID-19 pandemic, which emerged in late 2019, was caused by the novel coronavirus SARS-CoV-2. The

virus exhibited rapid human-to-human transmission and quickly escalated into a worldwide crisis affecting a large part of the global population. While initially recognized as a respiratory illness, COVID-19 has been found to affect multiple organs including the lungs, heart, vascular system, and kidneys. In patients with severe COVID-19, as many as one-third developed AKI early during the pandemic,¹⁹¹ which was histologically characterized mostly by acute tubular injury.¹⁹² However, glomerular abnormalities were observed in a subset of individuals afflicted by COVID-19, and a growing number of patients with glomerular disease manifesting in temporal relation to SARS-CoV-2 infection have been reported.¹⁹³

By far, the most reported pattern of glomerular injury was collapsing glomerulopathy, in the setting of COVID-19 also referred to as “COVID-19–associated nephropathy (COVAN),” in analogy to HIVAN (Fig. 34.9). COVAN typically manifested with nephrotic-range proteinuria alongside AKI.^{194–197} Much as HIVAN described earlier, COVAN manifests almost solely among patients of African descent.¹⁹⁸ Patients were predominantly male, and the vast majority of those tested exhibited high-risk APOL1 genetic variants (G1 and G2).^{194,195} The underlying mechanisms of COVAN are believed to be linked to immune system dysregulation. Elevated levels of specific cytokines including interleukin-1 β , interleukin-6, interleukin-10, and IFN- γ are observed during SARS-CoV-2 infection and could cause podocyte damage.¹⁹⁹ Direct viral infection of kidney cells has so far not been proven in COVAN, in contrast to HIVAN. The prognosis of affected patients was rather poor. About 56% of patients described in a systematic review suffered from persistent renal dysfunction, and 37% required definitive kidney replacement therapy.¹⁹³

Numerous case reports and case series have described a variety of other glomerular diseases manifesting in temporal association with SARS-CoV-2 infection, such as noncollapsing primary FSGS or minimal change disease, membranous nephropathy, ANCA-associated vasculitis, antglomerular basement membrane disease, and IgA

nephropathy.¹⁹³ However, whereas a causal role for SARS-CoV-2 appears likely in the case of COVAN, given the large number of reported cases and a plausible pathophysiologic mechanism, this remains questionable for all other glomerular diseases reported after COVID-19 so far. Relative frequencies of these glomerulopathies among patients with preceding SARS-CoV-2 infection were mostly not different from a historical biopsy cohort,¹⁹⁴ and their population-level incidence did not increase during the COVID-19 pandemic nor did it temporally correlate with COVID-19 incidence numbers in the preceding month during the pandemic.²⁰⁰ Thus the temporal association of the glomerular disease with COVID-19 may have arisen by chance in most reported cases.

Another important issue related to the COVID-19 pandemic is whether SARS-CoV-2 vaccinations themselves may induce glomerular disease. The rapid development of highly efficient vaccines, mostly based on the novel mRNA vaccine technology, has been instrumental in the fight against the SARS-CoV-2 pandemic. Although these vaccines have been tested in large-scale randomized controlled clinical trials with excellent safety profiles,^{201,202} numerous rare adverse events have been reported during the worldwide massive immunization campaigns. Among these reports were various types of glomerular diseases.^{203,204} However, adequately controlled studies did not find an increased risk for de novo glomerular disease after vaccination. Population-based incidence of glomerular diseases did not increase during or following the vaccination campaign,²⁰⁵ and neither a case-cohort design²⁰⁵ nor a self-controlled case series²⁰⁶ could detect an increased incidence of de novo glomerular disease following SARS-CoV-2 vaccination. The latter study²⁰⁶ also did not find an increased incidence of glomerular disease relapses after vaccination, although another cohort study found an increased risk of relapse after the second, but not the first, vaccine dose.²⁰⁷ Overall, it is likely that many cases of new-onset glomerular diseases, as well as relapses, were preceded by vaccination just by chance, given the vast amount of vaccines administered over a short period of time. This may explain the majority, if not all, cases of glomerular disease manifesting after vaccination.

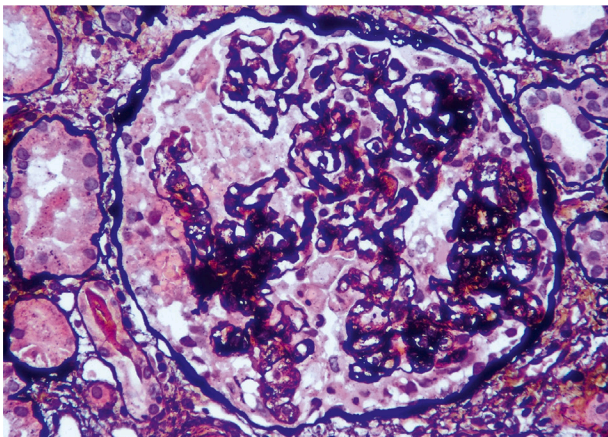


Fig. 34.9 Histology of a glomerulus from a patient with SARS-CoV2–associated nephropathy. The histopathology (Jones methenamine silver stain) in COVID-19–related glomerulonephritis shows segmental collapse of the glomerular tuft and proliferation of visceral epithelial cells. (Image kindly provided by Dr. Anila Abraham, Nephropathologist, Renopath, Chennai, India.)

PARASITIC INFECTIONS AND GLOMERULONEPHRITIS

In tropical regions and socioeconomically deprived communities, chronic parasitic infections are a common cause of CKD and kidney failure. Glomerulonephritis secondary to malaria, schistosomiasis, and filariasis are primarily mediated by immune complex formation and can coexist with other infections.

MALARIAL GLOMERULONEPHRITIS

Malaria, the most common endemic disease, is caused by plasmodium species transmitted through Anopheles female mosquito. The classical renal manifestation of malaria is acute tubular injury and obstruction due to hemoglobinuria, occurring in up to 59% of severe malaria.²⁰⁸ Less often, glomerular immune complex deposition leads to malarial glomerulonephritis. *Plasmodium falciparum* and *Plasmodium*

vivax contribute to malarial glomerulonephritis, yet the exact incidence is unclear. Affected individuals present with reversible AKI or chronic progressive kidney disease.^{103,209} Clinical manifestations include proteinuria, microscopic hematuria, renal dysfunction, and occasionally low-complement levels with underlying membranoproliferative or mesangioproliferative glomerulonephritis, minimal change disease, or collapsing glomerulopathy. Chronic infection by *Plasmodium malariae*, *vivax*, or *ovale* can lead to progressive glomerulonephritis known as “quartan malarial nephropathy,” which is now a rare cause for nephrotic syndrome and kidney dysfunction in children. Histopathologic lesions are graded by severity and include mild focal and segmental lesions (stage 1), moderate focal and segmental lesions (stage 2), diffuse or segmental lesions with interstitial and tubular changes (stage 3), and marked sclerosis and interstitial fibrosis and tubular atrophy (stage 4).

In the scenario of imported malaria seen in travelers of African ancestry, *P. falciparum* infection could be a trigger for FSGS, especially in those with genetic susceptibility or preexisting viral infection.²¹⁰ Recovery of glomerulonephritis is evident in nearly two thirds of cases of *P. falciparum* or *vivax*. Malarial glomerulonephritis should be treated with an appropriate regimen of antiparasitic therapy to eradicate the infection. Eradication does not always result in recovery of glomerulonephritis and progression may be observed within 3 to 5 years from primary infection. There is no indication for corticosteroids or immunosuppressive therapies for malarial glomerulonephritis.¹⁰³

SCHISTOSOMAL GLOMERULONEPHRITIS

Schistosomiasis is a tropical disease caused by trematodes that occurs across the regions of South America, Africa, Middle East, and Southeast Asia. The incidence of glomerulopathy secondary to *Schistosoma mansoni* (hepatosplenic form) and *Schistosoma haematobium* is around 12% to 50%.^{42,43} Young males are more commonly affected. Circulating schistosomal antigens deposit within the glomeruli, triggering immune complex formation and leading to a spectrum of pathologic glomerular lesions. Underlying histopathology patterns have been classified as class I (mesangial proliferative); class II (proliferative exudative lesions); class III (membranoproliferative lesions); class IV (FSGS); class V (amyloidosis); and class VI (cryoglobulinemia associated with coinfection with hepatitis C). Recently, an association of PLA2R-positive membranous nephropathy with schistosomal glomerulopathy has been reported.^{103,211}

Clinical manifestations range from asymptomatic proteinuria, hematuria, and nephritic syndrome to progressive kidney failure. Eosinophilia, hypergammaglobulinemia, and hypocomplementemia are common. Kidney biopsy is indicated in the presence of overt or progressive kidney disease or proteinuria >1 g/day, hypocomplementemia, hematuria, and reduced kidney function. A definitive diagnosis of schistosomal glomerulonephritis is made by identifying parasitic antigens via immunostaining in glomeruli.⁴³ The Schistosomiasis Consortium for Operational Research and Evaluation has invested efforts in developing various diagnostic assays including the urine-circulating anodic antigen test as a screening test for schistosomiasis in endemic

areas.²¹² Biomarkers of podocyte injury are being studied to address early detection of glomerular injury due to schistosomiasis.²¹³ Early therapy with praziquantel/oxamniquine can prevent development or progression of kidney disease. Once kidney disease is established, class I and II lesions respond to antiparasitic therapy or resolve spontaneously. However, proliferative lesions do not respond well to antiparasitic therapy or immunosuppression and can progress to kidney failure. Class VI lesions or coinfection with hepatitis C may require treatment with corticosteroids.¹⁰³

FILARIAL GLOMERULONEPHRITIS

In areas of poor vector control and limited health care facilities, filarial infections like *Loa loa*, *Onchocerca volvulus*, *Wuchereria bancrofti*, and *Brugia malayanica* occasionally present with proteinuria and hematuria. Kidney biopsies reveal mesangial proliferative or membranoproliferative glomerulonephritis, minimal change disease, collapsing-variant FSGS, or chronic sclerosing glomerulonephritis with or without evidence of microfilariae within arterioles, glomeruli, or interstitium (Fig. 34.10).²¹⁴

Filarial glomerulonephritis should be treated with adequate antiparasitic therapy to eradicate the organism. Antifilarial therapy can reduce milder forms of proteinuria. Treatment with diethylcarbamazine or ivermectin can worsen proteinuria or kidney function due to exacerbated immune response to antigen release by the dead parasite. Occasionally, therapeutic apheresis has been tried to reduce microfilarial load and antigen release before antifilarial therapy.^{103,215}

OUTLOOK

Ongoing research and advancements in our understanding of IRGNs hold promise for improved treatment strategies. As the medical community continues to investigate the precise mechanisms driving these kidney diseases, we can anticipate

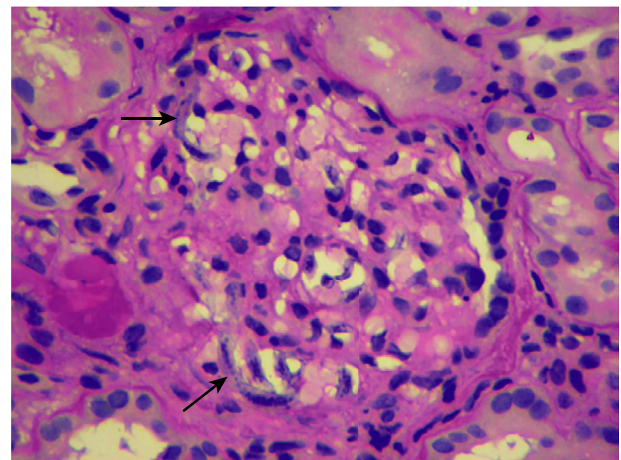


Fig. 34.10 Histology of a glomerulus from a patient with filarial glomerulonephritis. Light microscopy (PAS stain) of a kidney biopsy specimen with black arrows showing slender tube-like structures with parallel rows of nuclei representing microfilariae trapped within the capillaries of the glomerulus.²¹⁶ (Permission from Dr. Anila Abraham, Nephropathologist, Renopath, Chennai, India.)

the development of more tailored therapies that address specific infection-related triggers. A potentially promising avenue of intervention lies in the inhibition of specific parts of the complement system. Presently, a multitude of novel therapeutic agents is under development, offering the potential to selectively target and modulate particular components of the complement system in the future. In low-resource settings, a comprehensive approach that combines public health preventive measures with early diagnosis and affordable treatment options is essential to alleviate the burden of IRGNs in vulnerable populations. Collaboration among nephrologists, infectious disease specialists, basic researchers, and epidemiologists will be vital in shaping the future of care for IRGNs, offering hope for more effective management and better outcomes for our patients.

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